




Dynamic Programming

→ i/p → "n" post, "k" color

Condⁿ → not more than
adjacent post has same color

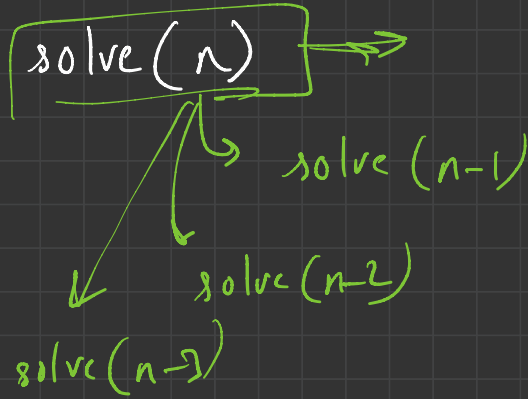
k=3

R G B

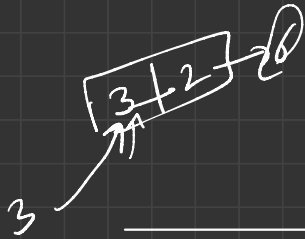
5 post

R	R	R	G	B
X	X	X	X	X
R	G	B	B	R

solve (n, k)



R.R = ?



$n=2$

x x

$K=3$ RGB

x x x
x x x
x x x

$n=4$

same

RR [3]
Gg ↓
BB [K]

RG BK
RB Bg
GR GB
GB

$K \cdot (K-1)$

9
↑

6
 $9 \times 2 = 18$

18

$24 \times (K-1)$
48

66

24

xxx
R G g
R B b
G R R
G B B
B R R
B g g → 6

Solve(2) → 9

$$\begin{aligned} & \hookrightarrow K + K \cdot (K-1) \\ & \rightarrow = K^2 \end{aligned}$$

diff

$R \boxed{GB}$

K

$\boxed{RB} \rightarrow K-1$
 $GB \rightarrow K-1$
 BB
 RG
 RB
 GR
 GB
 BR
 BG

} $\rightarrow K-1$

$\boxed{n=4}$

$\rightarrow$ 1st way $\rightarrow$ last 2 post $\rightarrow$ diff

2nd way $\rightarrow$ last 2 post $\rightarrow$ same

$n=4$

X X X X

same

diff

$n-2$

~~XX~~

RG

RB

BR

BB

BR

BB

RR

RR

BB

RRRR
RRBB

$$\text{solve}(n) \rightarrow \text{solve}(n-2) * (k-1) + \text{solve}(n-1) * (k-1)$$

$n=3$
X X X X

RG

RR

RB

RRR

RRB

RRR

RRR

RR

RR

RR

RR

RR

RR

RR

} k

R.R

{/

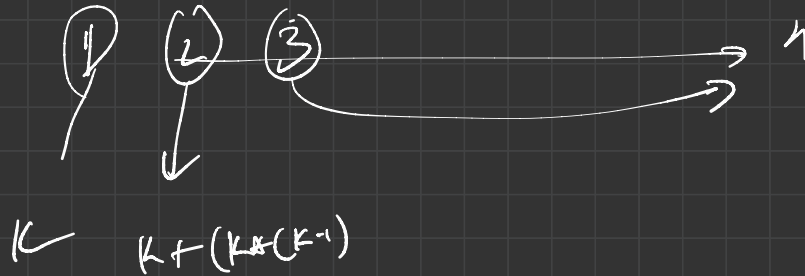
$$\text{solve}(n) = \text{solve}(n-2) * (k-1)$$

same

+

$$\text{solve}(n-1) * (k+1)$$

diff

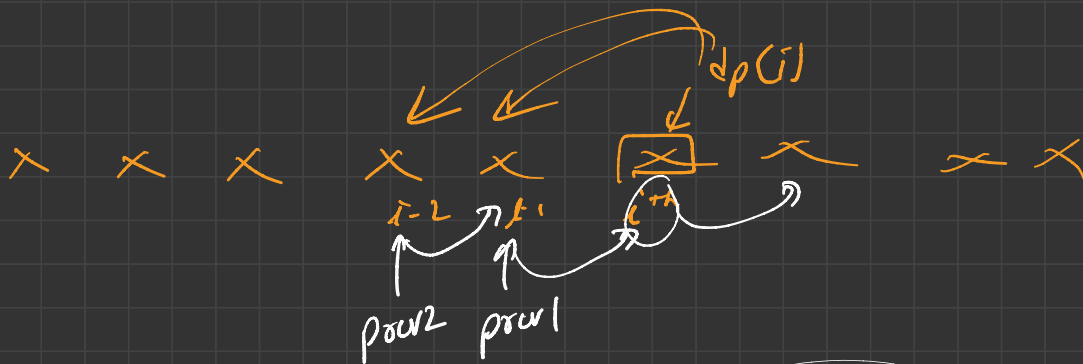


Rec $\rightarrow$ expo

Rec + Memo $\rightarrow$ T.C $\rightarrow O(N)$
 S.C $\rightarrow O(N) + O(N)$

Tab $\rightarrow$ T.C $\rightarrow O(N)$
 S.C $\rightarrow O(N) \xrightarrow{\text{S.O}} O(1)$

$dp[i] \rightarrow dp[i-1]$
 $dp[i] \rightarrow dp[i-2]$



$ans = prev1, prev2$

$prev2 = prev1$

$prev1 = ans$

